

1. which optimization produced the greatest improvement?

Ans: The scalability enhancement produced the greatest improvement. By optimizing thread management and improving connections handling, the chat server was able to support up to 10 concurrent clients without crashing. This significantly improved the application's performance, responsiveness, and stability under higher workloads.

2. why is scalability improvement?

Ans: Scalability is important because it enables an application to support an increasing number of users without degrading performance. A scalable system efficiently utilizes available resources, maintains stable communication, and provides consistent service even when multiple clients connect simultaneously.

3. what reliability issues did you discover?

Ans: The main reliability issues discovered were unexpected client disconnections, socket resource leaks, connection timeouts, and unhandled exceptions. These problems could cause application instability or server crashes. They were addressed by implementing automatic reconnection, proper resource cleanup, timeout handling, and improved exception handling.

4. which optimization was the most difficult?

Ans: The most difficult optimization was implementing automatic reconnection while preventing duplicate sessions. It required careful management of client states, proper synchronization between threads, and secure handling of reconnecting users without affecting existing active connections.

5. what additional improvements are required to support 100 users?

Ans: To support 100 users, the application should adopt asynchronous or event-driven networking. Instead of creating one thread per client. Additional improvements include using a database instead of CSV files, implementing load balancing, enabling TLS encryption, optimizing message broadcasting, improving memory management, and deploying the server on more powerful hardware or cloud infrastructure.